

CDIS: Coupling Compression Ratio and Token Entropy for Training-Free Instruction Selection

Anonymous submission

Abstract

Identifying high-quality instruction-response pairs is a critical bottleneck in instruction tuning. Current methods either incur substantial API costs through model-based scoring, or rely on single compression signals that cannot separate useful structure from degenerate repetition. We propose CDIS (Compression-Driven Instruction Selection), a CPU-only, ultra-low-cost framework that requires no model inference. CDIS couples Compression Ratio (CR) with Token Entropy (TE) over the full formatted dialogue turn: CR captures structural redundancy via lossless compression; TE provides complementary lexical diversity via Shannon entropy; together they address the compressibility ambiguity in instruction data. On Gemma-2-9B and Qwen2.5-7B, CDIS-1K is competitive with or exceeds several 9K-scale API- and model-dependent baselines using only 1,000 selected samples. Ablation studies confirm dual-metric necessity and show that uniform cluster quotas can impose a diversity tax under small budgets.

Introduction

Instruction tuning has become a standard paradigm for aligning Large Language Models (LLMs) with human intent (Wei et al. 2022; Ouyang et al. 2022), and a growing body of work shows that data quality, rather than quantity, shapes alignment outcomes (Zhou et al. 2023a; Gunasekar et al. 2023). Practical data selection has bifurcated into two costly regimes. LLM-based scoring methods such as AlpaGasus (Chen et al. 2024) and DEITA (Liu et al. 2024b) trade API cost for quality scores, while influence-based methods such as LESS (Xia et al. 2024) trade gradient computation for targeted capability induction. These approaches make high-quality selection effective but costly; diversity-preserving variants additionally rely on embedding-space clustering, whose uniform quotas can be brittle under very small budgets.

We propose CDIS (Compression-Driven Instruction Selection), a forward-free, ultra-low-cost framework that operates entirely on the CPU. CDIS scores each sample by the product of two complementary information-theoretic signals: Compression Ratio (CR), which captures structural redundancy via lossless compression, and Token Entropy (TE),

which captures lexical diversity via Shannon entropy over subword tokens. This dual signal addresses the compressibility ambiguity of single-metric compression methods: useful reasoning traces, code, and tables can be highly compressible, but so can degenerate templates or padding artifacts. Selection completes in minutes on commodity hardware (Table 6). Our contributions are: (i) the CDIS dual-signal selector; (ii) empirical quantification of the diversity tax, showing that removing K-Means clustering yields consistent gains under small-budget settings in our experiments; and (iii) evidence that 1,000 CDIS-selected samples are competitive with or exceed several 9K-scale API- and model-dependent baselines on two model families.

Related Work

Model and Influence-Based Data Selection

LLM-based scoring methods such as AlpaGasus (Chen et al. 2024) and DEITA (Liu et al. 2024b) use external teacher models (e.g., ChatGPT) to score and rank instruction samples, reducing 52K candidates to 9K–10K high-quality subsets. While effective, these approaches incur substantial API costs, introduce proxy-model domain bias, and struggle to scale to million-scale corpora. Influence-based methods such as LESS (Xia et al. 2024) estimate each sample’s impact on downstream performance via gradient similarity, achieving targeted capability induction at the cost of warmup training and backward-pass computation. SelectIT (Liu et al. 2024a) leverages the base LLM’s own uncertainty-aware self-reflection for ranking, eliminating external labels but requiring full-corpus model inference. Superfiltering (Li et al. 2024a) and IFD (Li et al. 2024b) reduce filtering cost through smaller proxy models or difficulty scores, but inherit the metric assumptions and bias of their proxy. All these methods share a common bottleneck: they require model forward passes or API queries during selection, limiting their practicality for large-scale, cost-sensitive data curation.

Diversity-Driven Clustering

Maintaining semantic diversity is widely considered essential in data selection. QDIT (Bukharin et al. 2023) explicitly optimizes a quality-diversity objective using facility-location-style submodular optimization over embeddings. CaR (Ge et al. 2024) combines expert-aligned quality scor-

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

ing with clustering for diversity-preserved selection. Cluster-Clip (Shao et al. 2024) uses clustering to rebalance frequent versus rare corpus regions during training. #InsTag (Lu et al. 2023) uses fine-grained instruction tagging to estimate diversity and complexity. Dataset Quantization (Zhou et al. 2023b) compresses datasets via embedding-based sampling for generic coverage.

These methods incur substantial computational overhead: embedding extraction for all samples, pairwise similarity computation, and combinatorial subset optimization. Our experiments indicate that explicit diversity enforcement through clustering can lower downstream performance in our setting, a phenomenon we term the *diversity tax* (Section). Recent information-gain selectors such as MIG (Chen et al. 2025), which models the semantic space via a label graph rather than K-Means, sit closer to compression-based selection and are included in our main comparison.

Compression-Based and Proxy-Based Data Curation

A recent line of work applies information-theoretic compression signals to data curation. The Entropy Law (Yin et al. 2024) links LLM performance to dataset compression ratio and first-epoch loss, proposing ZIP, a multi-stage greedy selector that iteratively prefers low-compression-ratio subsets. ZIP-FIT (Obbad et al. 2024) extends this to task alignment using gzip-based compression distance between source examples and target-task samples. Deletang et al. (2024) establish the equivalence between language modeling and compression, providing conceptual justification for compression-derived data signals, and Jiang et al. (2022) show that normalized compression distance with gzip plus kNN classifies text without learned parameters. Proxy-based scoring methods such as Superfiltering (Li et al. 2024a) and IFD (Li et al. 2024b) reduce filtering cost through smaller proxy models or difficulty scores, inheriting the assumptions of their proxy rather than the cost of a full LLM.

While compression-based methods are attractive for their CPU-only operation, existing approaches rely on a single compression-derived signal. ZIP’s greedy iterative selection over compression alone is vulnerable to a compressibility ambiguity: repeated artifacts, boilerplate, and other degenerate regularities can look highly compressible, while compression by itself does not test whether that structure is linguistically useful. CDIS addresses this by coupling CR with token entropy (TE), which discriminates between high-CR reasoning samples (structured knowledge, chain-of-thought) and high-CR artifacts (Unicode padding, repeated templates, encoding corruption). This dual-signal design, combined with a single-pass ranking rather than greedy iteration, gives both robustness and empirical gains.

Token-Level Statistics and Scaling Laws

Token-level quality signals are widely used in pretraining data curation. CCNet (Wenzek et al. 2020) filters Common Crawl by perplexity scoring. RefinedWeb (Penedo et al. 2023) demonstrates that aggressive filtering of web text can beat curated mixtures. Deduplication (Lee et al. 2022) shows

that removing repeated text improves both capability and evaluation validity.

The broader motivation for data selection comes from evidence that more data is not monotonically better. LIMA (Zhou et al. 2023a) demonstrates that a 65B model can become a strong chat model with only 1,000 carefully curated examples. Chinchilla (Hoffmann et al. 2022) establishes that model size and token budget must be balanced. Dataset Cartography (Swayamdipta et al. 2020) maps examples by training dynamics, showing that different regions contribute differently to generalization. Data Mixing Laws (Ye et al. 2025) demonstrates quantitative laws governing domain-mixture proportions. CDIS contributes to this evidence by providing an automatic, model-free selector that identifies a compact, information-dense core—extending LIMA’s manual curation insight into a reproducible selection framework.

Theoretical Framework

Submodular Maximization and the Compression Paradox

Given a candidate dataset $D = \{x_1, x_2, \dots, x_N\}$ of N instruction-response pairs formatted as dialogue sequences (Tables 1–2), data selection is modeled as a submodular maximization problem:

$$S^* = \arg \max_{S \subset D, |S|=k} \left(\sum_{x \in S} \mathcal{I}(x) - \lambda \mathcal{R}(S) \right) \quad (1)$$

where $\mathcal{I}(x)$ denotes intrinsic structural utility, $\mathcal{R}(S)$ penalizes within-subset redundancy, and λ balances the two terms. Existing approaches approximate $\mathcal{I}(x)$ through model forward passes, gradient similarity, or API queries. CDIS estimates $\mathcal{I}(x)$ entirely from physical compression and token-statistical features of the text sequence, requiring no model inference.

To quantify $\mathcal{I}(x_i)$ at low cost, we appeal to the Minimum Description Length (MDL) principle. Let $B(x_i)$ and $C(x_i)$ be the UTF-8 byte length and zstd-compressed length (level 3) of x_i . The compression ratio $CR_i = B(x_i)/C(x_i)$ captures structural compressibility: high CR_i indicates extensive repetitive patterns. In instruction data, this gives rise to the *compression paradox*—high compressibility simultaneously characterizes high-quality structured data (mathematical chain-of-thought, code with regular syntax) and low-quality noise (repeated Unicode fill characters). A single compression metric cannot disambiguate the two.

Token entropy resolves the ambiguity. Under the Gemma tokenizer, let V_i be the unique-token set of x_i with empirical probability $P_i(v)$. The Shannon entropy $TE_i = -\sum_{v \in V_i} P_i(v) \log_2 P_i(v)$ is large for diverse vocabulary and small for sharply peaked distributions. Meaningless artifacts exhibit narrow vocabulary, so $TE_i \rightarrow 0$; structured data retains lexical diversity alongside formatting regularity. Fusing normalized $\widetilde{CR}_i$ and $\widetilde{TE}_i$ multiplicatively into $\text{Score}_i = \widetilde{CR}_i \cdot \widetilde{TE}_i$ constructs a nonlinear joint criterion that collapses unless both signals are high. In the small- k regime studied here, the selected subset is far smaller

than the source pool, and utility differences among top-scoring samples can dominate redundancy penalties. We therefore use Score_i as a practical proxy for intrinsic utility, $\mathcal{I}(x_i) \approx \text{Score}_i$, and rank samples globally by this proxy. Under this approximation, the global top- k ranking (D4 and CDIS-1K) provides a simple greedy selection rule without requiring cluster-level redundancy control.

Selection Dynamics and the Diversity Tax

Traditional pipelines apply K-Means clustering with uniform per-cluster sampling quotas to minimize $\mathcal{R}(S)$. Under extreme selection budgets, this intervention inverts. Natural instruction corpora exhibit skewed quality distributions: the clustering space contains low-density regions $\mathcal{C}_{\text{low}}$ whose intra-cluster maximum score falls below the admission threshold of the global top- k subset $S_{\text{top-}k}$. When uniform cross-cluster quotas are enforced, the selector must draw from $\mathcal{C}_{\text{low}}$ to satisfy the partition constraint, displacing high-scoring samples from dense clusters with low-scoring substitutes. We define the resulting information loss as the *diversity tax*:

$$\Delta_{\text{tax}} = \sum_{x \in S_{\text{top-}k}} \text{Score}(x) - \sum_{x \in S_{\text{cluster}}} \text{Score}(x) > 0 \quad (2)$$

This expression makes the core tradeoff explicit: when $k \ll N$ and quality is skewed, the Δ_{tax} penalty from forced diversity can exceed the marginal benefit from redundancy reduction. The tax is not a fixed property of clustering but a function of the cluster-count-to-budget ratio. Under extreme budgets, global top- k selection (D4 and CDIS-1K) can provide a stronger approximation than uniform cluster quotas. Section provides empirical quantification.

Methodology

Preliminaries and Formatting

Let $D = \{(p_i, r_i)\}_{i=1}^N$ denote the raw instruction dataset, where p_i is the prompt and r_i the response. Data selection aims to identify a subset $S \subset D$ of size $k \ll N$ that maximizes downstream instruction-following performance after fine-tuning. We format each pair with the model’s official chat template into a context string $x_i = \text{Format}(p_i, r_i)$, so that both prompt and response quality are evaluated jointly and corrupted prompts common in crawled data are caught. The pipeline is summarized in Figure 1 and Table 2; notation is given in Table 1.

Information Measurement

The compression ratio $CR_i = B(x_i)/C(x_i)$ is measured with zstd at level 3 on the UTF-8 encoding of x_i . Values near 1 indicate nearly incompressible text with irregular structure; values far above 1 indicate text rich in repetitive patterns, templates, or structured formatting. Token entropy $TE_i = -\sum_{v \in V_i} P_i(v) \log_2 P_i(v)$ is computed from the Gemma tokenizer’s subword distribution over x_i and remains fixed across all selection experiments; the selected subset is then used to fine-tune both Gemma-2-9B and Qwen2.5-7B, making the selection pipeline model-agnostic. TE is large when vocabulary is varied and small when the token distribution is narrow. Both metrics are computed over the full

Table 1: Notation.

Symbol	Meaning
$D, (p_i, r_i)$	Dataset and prompt-response pair i
x_i	Formatted dialogue sequence for pair i
$B(x_i), C(x_i)$	UTF-8 and zstd-compressed byte lengths of x_i
CR_i, TE_i	Compression ratio and token entropy of x_i
$\widetilde{CR}_i, \widetilde{TE}_i$	Min-Max normalized CR and TE
Score_i	$\widetilde{CR}_i \cdot \widetilde{TE}_i$
$\mathcal{I}(x), \mathcal{R}(S)$	Intrinsic utility and redundancy penalty
Δ_{tax}	Diversity tax (quota-induced information loss)
μ_{CR}, σ_{CR}	Mean and std. dev. of CR over D
k	Target subset size

Table 2: CDIS pipeline overview.

Stage	Step	Metric	Detail
1	Format	Chat template	Gemma-2
2	CR	Compression Ratio	zstd (level=3)
3	TE	Token Entropy	Gemma tokenizer
4	Score	$CR_i \times TE_i$	Multiplicative fusion
5	Outlier Filter	σ -based	1.5σ on CR
6	Bucket Analysis	Score percentiles	Diagnostic
7	Final Sampling	Top- k by Score	No clustering

formatted dialogue turn, so corrupted prompts are caught and the instruction-response coupling is preserved.

Score Fusion and Selection Strategy

Following the theoretical reduction in Section , we compute $\text{Score}_i = \widetilde{CR}_i \cdot \widetilde{TE}_i$ as the proxy for $\mathcal{I}(x_i)$. Samples with $CR_i > \mu_{CR} + 1.5\sigma_{CR}$ are removed as outliers; the remainder are sorted by Score descending and the top- k are selected, directly approximating the greedy submodular solution under the small- k regime.

The CDIS Selection Algorithm

Algorithm 1 CDIS Selection

Require: Dataset $D = \{(p_i, r_i)\}_{i=1}^N$, tokenizer $\mathcal{T}$, target size k
Ensure: Subset $S \subseteq D$, $|S| = k$
1: **for** each $(p_i, r_i) \in D$ **do**
2: $x_i \leftarrow \text{Format}(p_i, r_i)$
3: $CR_i \leftarrow B(x_i)/C(x_i)$ {zstd, level 3}
4: $TE_i \leftarrow -\sum_v P_i(v) \log_2 P_i(v)$ {tokenizer $\mathcal{T}$ }
5: **end for**
6: $CR, TE \leftarrow \text{MinMaxNormalize}(CR, TE)$
7: **for** each i **do**
8: $\text{Score}_i \leftarrow \widetilde{CR}_i \cdot \widetilde{TE}_i$
9: **end for**
10: $D' \leftarrow \{i : CR_i \leq \mu_{CR} + 1.5\sigma_{CR}\}$
11: Sort D' by Score_i descending
12: $S \leftarrow \text{top-}k$ from D'
13: **return** S

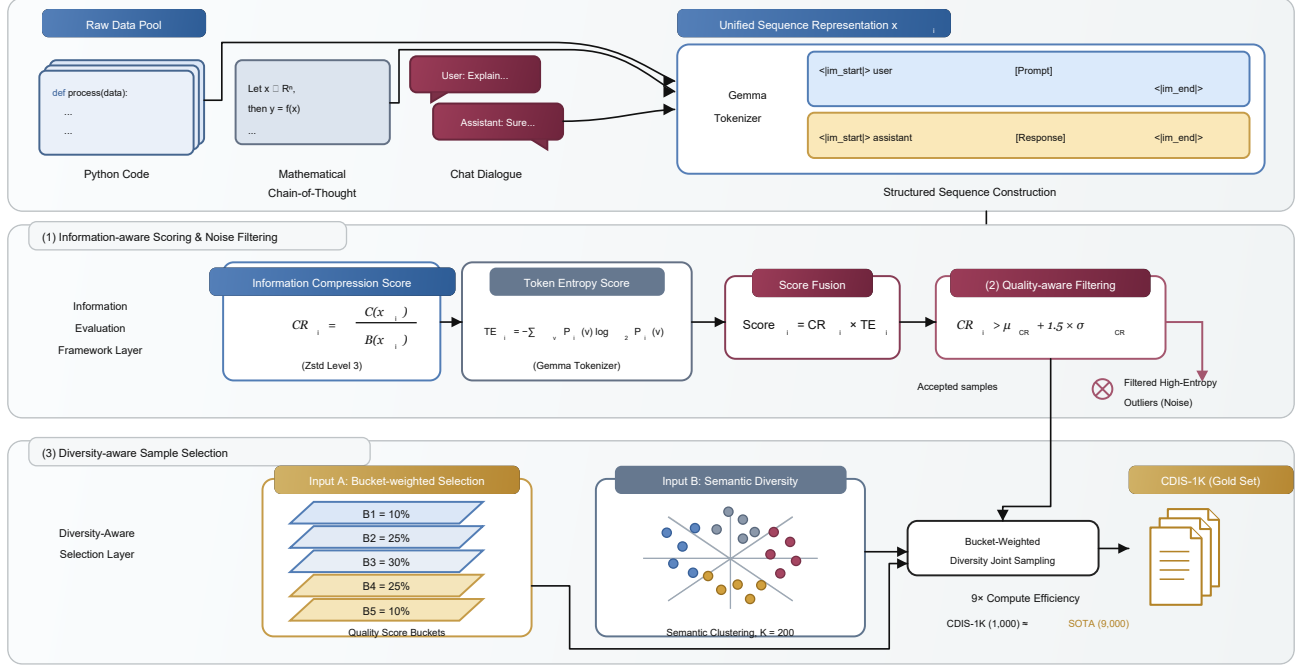


Figure 1: Overview of the CDIS pipeline. Top: heterogeneous instruction-response pairs are formatted into unified sequences x_i . Middle: dual-metric scoring ($CR \times TE$) with outlier filtering. Bottom: top- k selection by Score ranking.

Experiments

Experimental Setup

Datasets. We use the Alpaca-GPT4 dataset (52K instruction-response pairs) as the source corpus for all selection experiments. CDIS selection is performed once on this pool; the resulting subsets are used to fine-tune all downstream models.

Baselines. We compare against: Random selection; PPL (perplexity-based scoring under the base model); ZIP (Yin et al. 2024) (compression-based greedy selection, our re-implementation on the same pool); MIG (Chen et al. 2025) (information-gain selection via label-graph modeling); and AlphaGasus (Chen et al. 2024) (ChatGPT-based quality scoring with clustering). We additionally include Longest in the efficiency comparison as a simple CPU-only reference.

Benchmarks. We evaluate on ARC-Challenge (Clark et al. 2018) (25-shot, ‘acc_norm’) for reasoning, MMLU (Hendrycks et al. 2021) (5-shot, ‘acc’) for knowledge, and IFEval (Zhou et al. 2023c) (0-shot) for instruction following. IFEval uses the standard four metrics: instruction-level strict/loose (IL-S/IL-L) and prompt-level strict/loose (PL-S/PL-L). All evaluations use lm-evaluation-harness with chat-template rendering.

Implementation Details. CDIS uses zstd at compression level 3 for CR and the Gemma tokenizer for TE; both are computed on the full formatted dialogue turn. Samples with $CR_i > \mu_{CR} + 1.5\sigma_{CR}$ are removed as outliers. The remain-

ing samples are ranked by $\text{Score}_i = \widetilde{CR}_i \cdot \widetilde{TE}_i$ and the top- k are selected. Bucket stratification by score percentile is optional and treated as a tunable diagnostic.

Cross-tokenizer sensitivity. Table 3 reports TE correlations across four tokenizers. TE rankings are nearly identical across all subword tokenizers (Spearman $\rho \geq 0.92$, top-1K Jaccard 71.8–81.5% under identical CR). TE can thus be computed with any standard tokenizer without altering CDIS rankings, confirming that CDIS does not depend on a specific tokenizer.

Table 3: Top-1,000 selection stability across tokenizers.

Tokenizer pair	Spearman ρ	Jaccard @ 1K
Gemma BPE vs. Qwen BPE	0.94	77.3%
Gemma BPE vs. Word	0.92	71.8%
Qwen BPE vs. Word	0.98	81.5%
Gemma BPE vs. Char	0.62	20.7%

Cross-compressor sensitivity. Table 4 shows CR rankings are stable across compressors (pairwise Spearman $\rho \geq 0.989$, top-1K Jaccard $> 78\%$). zstd level has negligible impact ($\rho = 0.998$ between levels 1 and 19), confirming zstd-3 as a reasonable speed-compression tradeoff.

Training Details. We fine-tune Gemma-2-9B (Team et al. 2024) and Qwen2.5-7B-Instruct with QLoRA (4-bit NF4, $r=64$, $\alpha=16$, dropout 0.05), effective batch size 32, learning rate 2×10^{-4} , one epoch.

Table 4: CR correlations across compressors (pairwise Spearman ρ).

	zstd-1	zstd-19	gzip-6	BZ2	LZMA
zstd-1	1	0.998	0.996	0.991	0.989
zstd-19		1	0.996	0.990	0.989
gzip-6			1	0.996	0.996
BZ2				1	0.996

Main Experimental Results

Overall performance with ultra-low data budget. Table 5 reports results on two model families. On Gemma-2-9B, CDIS-1K achieves the highest ARC-C (68.34%), surpassing all 9K-scale baselines including PPL-9K (67.41%), MIG (67.24%), AlpaGasus (67.15%), and ZIP (66.04%). On IFEval, CDIS-1K leads on both prompt-level metrics (PL-S 57.67%, PL-L 61.37%), exceeding AlpaGasus by 1.29 and 1.11 points, and is within 1.0 point on instruction-level metrics. On MMLU, CDIS-1K (70.30%) is 0.20 points behind AlpaGasus (70.50%), the strongest baseline.

Cross-model generalization. On Qwen2.5-7B-Instruct, CDIS-1K ties ZIP for best ARC-C (62.54%) and leads on MMLU (73.86%, second-best MIG at 73.11%). AlpaGasus achieves 61.86% ARC-C and 72.70% MMLU. The same CDIS-selected subset transfers across model families without modification, since the selection pipeline uses only physical compression and token statistics and requires no model-specific inference.

Computational efficiency. Table 6 compares selection cost. CDIS operates entirely on the CPU and selects 9K samples from the full 52K pool in approximately 3 minutes. In contrast, PPL requires GPU inference (30–45 minutes on a V100), AlpaGasus incurs API costs scaling with dataset size, and MIG, while GPU-based, completes in around 2 minutes. Among nontrivial quality-aware selectors, CDIS combines CPU-only operation with sub-5-minute wall-clock time and zero external API cost.

Insight: forced diversity can harm performance. A striking result is that the no-cluster variant D4 (68.00% ARC-C) outperforms the clustering-augmented D5 (Cluster-only, 66.72%) by 1.28 points. Under small selection budgets, uniform per-cluster sampling quotas force the selector to draw from low-density, low-quality clusters, displacing higher-scoring samples from dense regions. This diversity tax is not a condemnation of clustering broadly, but a budget-dependent phenomenon explored in Section .

Ablation Studies and Analysis

Core component ablation. Table 7 isolates the contribution of each CDIS component via controlled removal from the D4 baseline. Removing either signal causes a clear drop: CR-only (D2, 65.27%) and TE-only (D3, 65.87%) both underperform random selection (65.78%), confirming that neither signal alone suffices. Removing the Score ranking while keeping clustering (D5, 66.72%) costs 1.28 points. Length-matched random selection (D1, 65.78%) performs compara-

bly to pure random, indicating that length is not the driving factor.

Fusion operator comparison. Table 8 reports alternative fusion operators. The max operator (67.75%) approaches multiplicative fusion (D4, 68.00%), while a learned convex combination (66.55%) underperforms both, suggesting that simple multiplicative fusion already captures the essential interaction between CR and TE without over-parameterization.

Scaling behavior. Increasing the selection budget from 1K to 5K does not yield monotonic ARC-C improvement: CDIS-1K (68.34%) > CDIS-3K (66.72%) > CDIS-5K (66.30%). Each additional batch beyond the top 1,000 introduces progressively lower-ranked samples, consistent with the alignment tax discussed in Section .

Discussion

Diversity Tax

Clustering is widely regarded as essential for preserving semantic diversity in data selection (Chen et al. 2024; Liu et al. 2024b; Ge et al. 2024). In our setting, however, the clustering-augmented configuration (D5: Cluster-only, 66.72%) underperforms the no-cluster D4 variant (68.00% ARC-C), a gap we refer to as the *diversity tax*. We emphasize that this result does not indict clustering as a technique; rather, it reveals a structural tension between uniform cross-cluster quotas and extreme selection budgets.

The mechanism is statistical. Real-world SFT corpora such as Alpaca-GPT4 exhibit a highly skewed quality distribution across semantic regions: a small number of dense, high-quality clusters concentrate most informative samples, while a long tail of sparse, low-quality clusters carries few useful examples. Under a standard K-Means pipeline with uniform per-cluster quotas, every cluster—regardless of its density or quality—must contribute an equal share of samples. When the total selection budget is small relative to the number of clusters (e.g., $K=200$ clusters competing for a 1,000-sample budget), the per-cluster allocation drops to only a handful of samples. The sampler is forced to fill these quotas by reaching into low-density peripheral clusters, pulling in noisy or uninformative samples that the global Score ranking would have naturally excluded. The result is quality dilution: high-scoring samples in dense clusters are displaced by low-scoring samples from sparse clusters, purely to satisfy the quota constraint.

This account implies that the diversity tax is not a fixed property of clustering, but a function of the cluster-count-to-budget ratio. Under larger budgets or fewer clusters, the per-cluster quota is large enough to draw exclusively from each cluster’s high-scoring region, and the tax should diminish. Conversely, under extreme budgets, any method that enforces uniform cross-partition sampling will face the same tension. We therefore regard the diversity tax as a budget-sensitive phenomenon rather than a categorical critique of diversity enforcement, and we note that milder relaxations such as density-weighted quotas or score-space stratification may recover clustering’s benefits without incurring the tax.

Figure 2 visualizes this trade-off across all evaluated methods. CDIS-1K lies on the Pareto frontier, achieving the best

Table 5: Main evaluation results. Best per column within each model section are **bold**; second-best are underlined.

Method	Size	ARC-C	MMLU	IL-S	IL-L	PL-S	PL-L
<i>Gemma-2-9B</i>							
CDIS-1K	1K	68.34	<u>70.30</u>	<u>66.07</u>	<u>69.54</u>	57.67	61.37
PPL-9K	9K	<u>67.41</u>	70.15	59.83	63.67	47.50	51.57
MIG (Chen et al. 2025)	9K	67.24	69.88	58.15	62.35	47.69	52.50
AlpaGasus (Chen et al. 2024)	9K	67.15	70.50	67.03	70.02	<u>56.38</u>	<u>60.26</u>
ZIP (Yin et al. 2024)	9K	66.04	69.65	58.15	61.87	46.77	50.28
Random	9K	65.78	69.85	60.67	64.39	49.54	52.87
<i>Qwen2.5-7B-Instruct</i>							
CDIS-1K	1K	62.54	73.86	—	—	—	—
AlpaGasus (Chen et al. 2024)	9K	61.86	72.70	—	—	—	—
ZIP (Yin et al. 2024)	9K	62.54	72.73	—	—	—	—
PPL-9K	9K	61.01	72.64	—	—	—	—
MIG (Chen et al. 2025)	9K	60.32	<u>73.11</u>	—	—	—	—

Table 6: Selection efficiency comparison. *Time* denotes approximate wall-clock time to select 9K samples from a 52K candidate pool.

Method	Hardware	Fwd. Pass	API Cost	Time
Random	CPU	No	No	<1 s
Longest	CPU	No	No	<1 s
CDIS	CPU	No	No	~3 min
PPL	V100	Yes	No	30–45 min
MIG	V100	No	No	~2 min
AlpaGasus	API	No	Yes	hours

Table 7: Component ablation: ARC-Challenge drop relative to D4: No-cluster (68.00%).

Removed	Variant	ARC-C Drop
—	D4: No-cluster	0.00 (baseline)
Clustering	D5: Cluster-only	−1.28
TE	D3: TE-only	−2.13
CR	D2: CR-only	−2.73
Length	D1: Length-Matched	−2.22

ARC-Challenge accuracy with only 1,000 samples. No other method dominates it.

Scaling and the Alignment Tax

Figure 3 plots CDIS performance as the selection budget grows from 1K to 5K. CDIS-1K attains the peak ARC-C score (68.34%), and each subsequent increase in data volume lowers performance. We call this decline the *alignment tax*: a penalty for moving beyond the highest-ranked core under a fixed training recipe. The trend is consistent with the LIMA observation (Zhou et al. 2023a) that alignment can saturate with a small, high-quality core, after which additional samples may dilute rather than enhance the training signal.

The ablation results further quantify the alignment tax. Moving from CDIS-1K (68.34%) to CDIS-3K (66.72%) to CDIS-5K (66.30%), ARC-Challenge performance declines.

Table 8: Full ablation, scaling, and fusion results on Gemma-2-9B.

Method	Size	ARC-C	MMLU	Type
<i>CDIS Configurations</i>				
D4: No-cluster	9K	68.00	70.28	Reference
CDIS 3K	3K	66.72	69.56	Scaling
CDIS 5K	5K	66.30	69.48	Scaling
<i>Component Ablation</i>				
D2: CR-only	9K	65.27	69.23	CR removed
D3: TE-only	9K	65.87	69.95	TE removed
D5: Cluster-only	9K	66.72	69.15	Score removed
D1: Len-Matched	9K	65.78	69.71	Length ctrl.
<i>Fusion Operator</i>				
Fusion (max)	9K	67.75	70.16	$\max(\overline{CR}, \overline{TE})$
Fusion (learned)	9K	66.55	69.72	$\alpha \overline{CR} + (1 - \alpha) \overline{TE}$

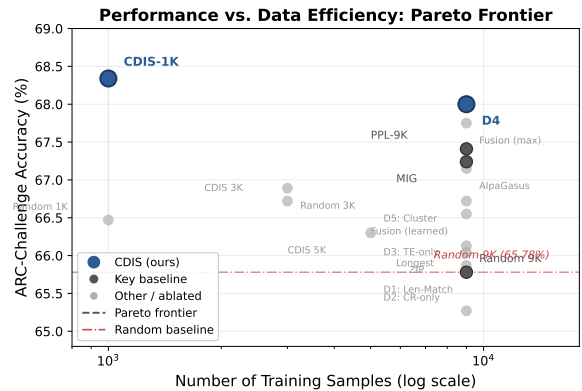


Figure 2: Performance–efficiency Pareto frontier on ARC-Challenge. CDIS-1K sits at the frontier with the best accuracy using the smallest data budget. The frontier demonstrates that data quality, not quantity, drives ARC-Challenge performance.

Each additional batch beyond the top 1,000 introduces lower-ranked samples under the CDIS score. This non-monotonic scaling suggests that alignment can saturate at a compact

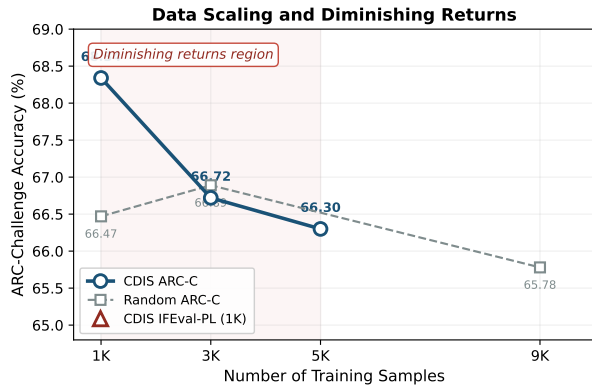


Figure 3: Data scaling and diminishing returns. CDIS ARC-Challenge accuracy declines from 1K to 5K while the random baseline remains flat; the shaded region marks the diminishing-returns zone discussed as the alignment tax.

Q1 at the **The Compression Paradox: Why CR Alone Cannot Disambiguate** diversity.

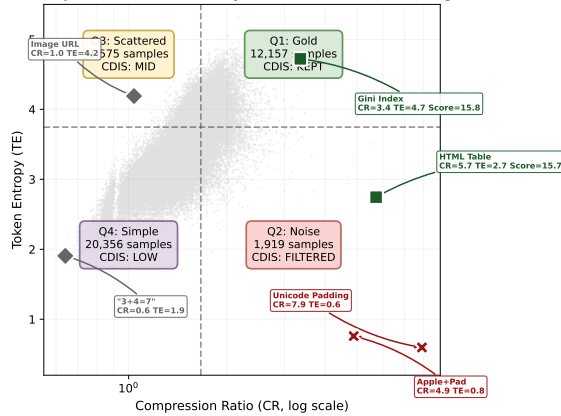


Figure 4: Four-quadrant decomposition of the CR-TE space. CDIS retains high-CR/high-TE samples (Q1), filters high-CR/low-TE artifacts (Q2), assigns mid scores to low-CR/high-TE samples (Q3), and deprioritizes low-CR/low-TE trivia (Q4). Dashed lines mark $\mu + 0.5\sigma$ thresholds.

information-capacity threshold under the evaluated training recipe.

The four-quadrant structure of the CR-TE space offers further insight.

Why does TE matter? A compression-only metric is vulnerable to what we call the *compressibility ambiguity*: high CR can reflect useful structure, but it can also reflect repeated templates, padding artifacts, or other degenerate regularities. Our CR-only variant (D2, 65.27%) underperforms even random selection (65.78%), consistent with this ambiguity.

TE mitigates it: degenerate repeated text can obtain high CR while retaining a narrow token distribution, whereas structured reasoning samples keep high TE because their vocabulary remains varied. Multiplicative fusion thus preserves high-CR, high-TE reasoning samples and filters high-CR, low-TE artifacts.

Limitations and Future Work

Several limitations bound the current study. All experiments use QLoRA fine-tuning with a single-epoch recipe; the in-

teraction between CDIS selection and training hyperparameters has not been explored. The bucket proportions and outlier threshold were set heuristically and treated as tunable settings. The current evaluation covers two model families and one source dataset; broader validation on larger models, additional corpora, and multi-turn dialogue data remains open. Promising future directions include exploring alternative compressors and their interaction with CR behavior, extending CDIS to larger model families and diverse data pools, and combining CDIS as a lightweight first-stage CPU filter with a second-stage model-based selector for coarse-to-fine curation at scale.

Conclusion

We proposed CDIS, a forward-free, CPU-only data selection framework that scores samples by the product of Compression Ratio and Token Entropy, capturing global structural redundancy and local vocabulary diversity without model inference. Our empirical analysis identifies a diversity tax in clustering-augmented selection, and shows that removing K-Means clustering yields consistent gains under small-budget settings in our experiments. With 1,000 samples, CDIS-1K is competitive with or exceeds several 9K-scale baselines on ARC-Challenge and IFEval. We suggest that SFT data selection can benefit from small, high-quality subsets selected via forward-free metrics, complementing rather than replacing diversity- or influence-driven methods.

References

- Bukharin, A.; Li, S.; Wang, Z.; Yang, J.; Yin, B.; Li, X.; Zhang, C.; Zhao, T.; and Jiang, H. 2023. Data Diversity Matters for Robust Instruction Tuning. *arXiv preprint arXiv:2311.14736*.
- Chen, L.; Li, S.; Yan, J.; Wang, H.; Gunaratna, K.; Yadav, V.; Tang, Z.; Srinivasan, V.; Zhou, T.; Huang, H.; and Jin, H. 2024. AlpaGasus: Training A Better Alpaca with Fewer Data. In *International Conference on Learning Representations (ICLR)*.
- Chen, Y.; Li, Y.; Hu, K.; Zerun, M.; Ye, H.; and Chen, K. 2025. MIG: Automatic Data Selection for Instruction Tuning by Maximizing Information Gain in Semantic Space. In *Findings of the Association for Computational Linguistics: ACL 2025*.
- Clark, P.; Cowhey, I.; Etzioni, O.; Khot, T.; Sabharwal, A.; Schoenick, C.; and Tafjord, O. 2018. Think you have Solved Question Answering? Try ARC, the A12 Reasoning Challenge. In *arXiv preprint arXiv:1803.05457*.
- Deletang, G.; Ruoss, A.; Duquenne, P.-A.; Catt, E.; Genewein, T.; Mattern, C.; Grau-Moya, J.; Wenliang, L. K.; Aitchison, M.; Orseau, L.; Hutter, M.; and Veness, J. 2024. Language Modeling Is Compression. In *International Conference on Learning Representations*.
- Ge, Y.; Liu, Y.; Hu, C.; Meng, W.; Tao, S.; Zhao, X.; Xia, M.; Li, Z.; Chen, B.; Yang, H.; Li, B.; Xiao, T.; and Zhu, J. 2024. Clustering and Ranking: Diversity-preserved Instruction Selection through Expert-aligned Quality Estimation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 464–478.

- Gunasekar, S.; Zhang, Y.; Aneja, J.; Mendes, C. C. T.; Giorno, A. D.; Gopi, S.; Javaheripi, M.; Kauffmann, P.; de Rosa, G.; Saarikivi, O.; Salim, A.; Shah, S.; Behl, H. S.; Wang, X.; Bubeck, S.; Eldan, R.; Kalai, A. T.; Lee, Y. T.; and Li, Y. 2023. Textbooks Are All You Need. *arXiv preprint arXiv:2306.11644*.
- Hendrycks, D.; Burns, C.; Basart, S.; Zou, A.; Mazeika, M.; Song, D.; and Steinhardt, J. 2021. Measuring Massive Multitask Language Understanding. *arXiv preprint arXiv:2009.03300*.
- Hoffmann, J.; Borgeaud, S.; Mensch, A.; Buchatskaya, E.; Cai, T.; Rutherford, E.; de Las Casas, D.; Hendricks, L. A.; Welbl, J.; Clark, A.; Hennigan, T.; Noland, E.; Millican, K.; van den Driessche, G.; Damoc, B.; Guy, A.; Osindero, S.; Simonyan, K.; Elsen, E.; Rae, J. W.; Vinyals, O.; and Sifre, L. 2022. Training Compute-Optimal Large Language Models. *arXiv preprint arXiv:2203.15556*.
- Jiang, Z.; Yang, M. Y. R.; Tsirlin, M.; Tang, R.; and Lin, J. 2022. Less is More: Parameter-Free Text Classification with Gzip. *arXiv preprint arXiv:2212.09410*.
- Lee, K.; Ippolito, D.; Nystrom, A.; Zhang, C.; Eck, D.; Callison-Burch, C.; and Carlini, N. 2022. Deduplicating Training Data Makes Language Models Better. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*.
- Li, M.; Zhang, Y.; He, S.; Li, Z.; Zhao, H.; Wang, J.; Cheng, N.; and Zhou, T. 2024a. Superfiltering: Weak-to-Strong Data Filtering for Fast Instruction-Tuning. *arXiv preprint arXiv:2402.00530*. ACL 2024.
- Li, M.; Zhang, Y.; Li, Z.; Chen, J.; Chen, L.; Cheng, N.; Wang, J.; Zhou, T.; and Xiao, J. 2024b. From Quantity to Quality: Boosting LLM Performance with Self-Guided Data Selection for Instruction Tuning. *arXiv preprint arXiv:2308.12032*. NAACL 2024.
- Liu, L.; Liu, X.; Wong, D. F.; Li, D.; Wang, Z.; Hu, B.; and Zhang, M. 2024a. SelectIT: Selective Instruction Tuning for LLMs via Uncertainty-Aware Self-Reflection. *arXiv preprint arXiv:2402.16705*. NeurIPS 2024.
- Liu, W.; Zeng, W.; He, K.; Jiang, Y.; and He, J. 2024b. What Makes Good Data for Alignment? A Comprehensive Study of Automatic Data Selection in Instruction Tuning. In *International Conference on Learning Representations (ICLR)*.
- Lu, K.; Yuan, H.; Yuan, Z.; Lin, R.; Lin, J.; Tan, C.; Zhou, C.; and Zhou, J. 2023. #InsTag: Instruction Tagging for Analyzing Supervised Fine-tuning of Large Language Models. *arXiv preprint arXiv:2308.07074*.
- Obbad, E.; Mlaoui, I.; Miranda, B.; Obbad, K.; Schaeffer, R.; Bedi, S.; and Koyejo, S. 2024. ZIP-FIT: Embedding-Free Data Selection via Compression-Based Alignment. Open-Review submission to ICLR 2025.
- Ouyang, L.; Wu, J.; Jiang, X.; Almeida, D.; Wainwright, C. L.; Mishkin, P.; Zhang, C.; Agarwal, S.; Slama, K.; Ray, A.; Schulman, J.; Hilton, J.; Kelton, F.; Miller, L.; Simens, M.; Askell, A.; Welinder, P.; Christiano, P.; Leike, J.; and Lowe, R. 2022. Training Language Models to Follow Instructions with Human Feedback. *arXiv preprint arXiv:2203.02155*.
- Penedo, G.; Malartic, Q.; Hesslow, D.; Cojocaru, R.; Cappelli, A.; Aobaidli, H.; Pannier, B.; Almazrouei, E.; and Lounay, J. 2023. The RefinedWeb Dataset for Falcon LLM: Outperforming Curated Corpora with Web Data, and Web Data Only. *arXiv preprint arXiv:2306.01116*.
- Shao, Y.; Li, L.; Fei, Z.; Yan, H.; Lin, D.; and Qiu, X. 2024. Balanced Data Sampling for Language Model Training with Clustering. In *Findings of the Association for Computational Linguistics: ACL 2024*, 14012–14023.
- Swayamdipta, S.; Schwartz, R.; Lourie, N.; Wang, Y.; Hajishirzi, H.; Smith, N. A.; and Choi, Y. 2020. Dataset Cartography: Mapping and Diagnosing Datasets with Training Dynamics. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 9275–9293.
- Team, G.; Riviere, M.; Pathak, S.; Sessa, P. G.; Hardin, C.; Bhupatiraju, S.; Hussenot, L.; Mesnard, T.; Shahriari, B.; Ramé, A.; Ferret, J.; Liu, P.; Kalai, A. T.; Hees, A.; Vieillard, N.; Dadashi, R.; Pietruszka, M.; Andreev, A.; Remy, A.; Grewe, D.; Genewein, T.; Wang, X.; Lazewska, M.; Thillai, N.; Kulizhskaya, Y.; Pinto, M.; Kirk, P.; Bai, L.; Ghahramani, Z.; Hadsell, R.; and Vinyals, O. 2024. Gemma 2: Improving Open Language Models at a Practical Size. *arXiv preprint arXiv:2408.00118*.
- Wei, J.; Bosma, M.; Zhao, V. Y.; Guu, K.; Yu, A. W.; Lester, B.; Du, N.; Dai, A. M.; and Le, Q. V. 2022. Finetuned Language Models Are Zero-Shot Learners. *arXiv preprint arXiv:2109.01652*.
- Wenzek, G.; Lachaux, M.-A.; Conneau, A.; Chaudhary, V.; Guzman, F.; Joulin, A.; and Grave, E. 2020. CCNet: Extracting High Quality Monolingual Datasets from Web Crawl Data. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 4003–4012.
- Xia, M.; Malladi, S.; Gururangan, S.; Arora, S.; and Chen, D. 2024. LESS: Selecting Influential Data for Targeted Instruction Tuning. In *Proceedings of the 41st International Conference on Machine Learning*.
- Ye, J.; Liu, P.; Sun, T.; Zhan, J.; Zhou, Y.; and Qiu, X. 2025. Data Mixing Laws: Optimizing Data Mixtures by Predicting Language Modeling Performance. *arXiv preprint arXiv:2403.16952*. ICLR 2025.
- Yin, M.; Wu, C.; Wang, Y.; Wang, H.; Guo, W.; Wang, Y.; Liu, Y.; Tang, R.; Lian, D.; and Chen, E. 2024. Entropy Law: The Story Behind Data Compression and LLM Performance. *arXiv preprint arXiv:2407.06645*.
- Zhou, C.; Liu, P.; Xu, P.; Iyer, S.; Sun, J.; Mao, Y.; Ma, X.; Efrat, A.; Yu, P.; Yu, L.; Zhang, S.; Ghosh, G.; Lewis, M.; Zettlemoyer, L.; and Levy, O. 2023a. LIMA: Less Is More for Alignment. *arXiv preprint arXiv:2305.11206*.
- Zhou, D.; Wang, K.; Gu, J.; Peng, X.; Lian, D.; Zhang, Y.; You, Y.; and Feng, J. 2023b. Dataset Quantization. *arXiv preprint arXiv:2308.10524*. ICCV 2023.
- Zhou, J.; Lu, T.; Mishra, S.; Brahma, S.; Basu, S.; Luan, Y.; Zhou, D.; and Hou, L. 2023c. Instruction-Following

Evaluation for Large Language Models. *arXiv preprint
arXiv:2311.07911*.